Implementation procedure and system placement

Repository: https://github.com/yerkebttt/municipal-air-quality-batch-ingestion

System description

A Python loader sits between the public UCI source file and MongoDB. A separate API or BI layer can later serve station/time queries to the city dashboard. Docker Compose starts MongoDB and the loader, which waits for the database health check. An initialisation script validates the shared fields and creates a station/timestamp index.

Procedure

The loader downloads the archive and reads the CSV files row by row. Each row becomes a station-hour document: missing values are omitted, unknown measurement columns are kept, and the source time is stored as Asia/Shanghai. Documents are written in batches of 1,000. Deterministic IDs and `$setOnInsert` upserts make a replay after a partial write safe, and transient database errors are retried up to five times with exponential backoff (MongoDB, n.d.-c).

Usage and checks

Clone the repository and run `docker compose up --build`. After ingestion, `db.measurements.countDocuments({})` should return 420,768, and a query by station and time should return the matching readings. Unit tests cover normalisation, missing values, future fields and batching.

Reference

MongoDB. (n.d.-c). *PyMongo bulk write operations*. https://www.mongodb.com/docs/languages/python/pymongo-driver/current/crud/bulk-write/